

Programme	B. Sc. Mathematics Honours			
Course Code	MAT6EJ305			
Course Title	TOPOLOGY OF METRIC SPACES			
Type of Course	Elective			
Semester	VI			
Academic Level	300 - 399			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	1. Fundamental Mathematics Concepts: Set, Functions, Logic 2. Real Analysis			
Course Summary	This course familiarises students with the basic tools and phenomenology of topology by introducing metric spaces as a generalisation of the familiar Euclidean spaces.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate understanding of fundamental concepts in metric spaces and basic examples of metric spaces.	U	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO2	To analyse and evaluate the basic topology of metric spaces, including open sets, closed sets, interior, closure, and boundary points	An	E	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO3	Demonstrate proficiency in applying concepts of convergence, completeness, and continuity in metric spaces, including understanding Cauchy sequences, completeness, and continuity of functions.	Ap	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook	Introduction to Topology and Modern Analysis, George F. Simmons, Krieger Publishing Company (1982) ISBN-0-89874-551-9		
Module	Unit	Content	Hrs (48+ 12)
I	Introduction to Metric Spaces		12
	1	Chapter 1 Section 5: Partitions and Equivalence Relations	
	2	Chapter 1 Section 6: Countable Sets	
	3	Chapter 1 Section 7: Uncountable Sets	
	4	Chapter 2 Section 9: The Definition and Some Examples (Topics up to and including Example 2)	
	5	Chapter 2 Section 9: The Definition and Some Examples (Topics from Example 3 onwards)	
II	Basic Topology of Metric Spaces		10
	6	Chapter 2 Section 10: Open Sets (Topics up to and including Theorem A)	
	7	Chapter 2 Section 10: Open Sets (Theorem B and Theorem C)	
	8	Chapter 2 Section 10: Open Sets (Topics from Theorem D onwards)	
	9	Chapter 2 Section 11: Closed Sets (Topics up to and including Theorem C)	
	10	Chapter 2 Section 11: Closed Sets (Topics from Theorem D onwards)	
III	Convergence, Completeness & Continuity		12
	11	Chapter 2 Section 12: Convergence, Completeness, and Baire's Theorem (Topics up to Theorem A)	
	12	Chapter 2 Section 12: Convergence, Completeness, and Baire's Theorem (Theorem A and Theorem B)	
	13	Chapter 2 Section 12: Convergence, Completeness, and Baire's Theorem (Topics from Theorem C onwards)	
	14	Chapter 2 Section 13: Continuous Mappings (Topics up to and including Theorem A)	
	15	Chapter 2 Section 13: Continuous Mappings (Theorem B and Theorem C)	
	16	Chapter 2 Section 13: Continuous Mappings (Topics from Theorem D onwards)	
IV	Special Classes of Metric Spaces		14
	17	Chapter 2 Section 14: Spaces of Continuous Functions (Topics up to First Lemma)	
	18	Chapter 2 Section 14: Spaces of Continuous Functions (First Lemma, Second Lemma)	
	19	Chapter 2 Section 14: Spaces of Continuous Functions (Topics from Theorem A onwards)	
	20	Chapter 2 Section 15: Euclidean and Unitary Spaces (Topics up to First Lemma)	
	21	Chapter 2 Section 15: Euclidean and Unitary Spaces (First Lemma, Second Lemma)	
	22	Chapter 2 Section 15: Euclidean and Unitary Spaces (Topics from Theorem A onwards)	
Compactness In Metric Spaces			

V (Open Ended)		The Heine-Borel Property Bolzano-Weierstrass Property Lebesgue's Covering Lemma Sequential Compactness Compactness – Open Cover Formulation Total Boundedness Compactness, Completeness & Total Boundedness Equicontinuity & the Arzela-Ascoli Theorem	12
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References:

1. Introduction to General Topology, K. D. Joshi, New Age International.
2. A First Course In Topology, James R. Munkres, Prentice Hall of India
3. Topology of Metric Spaces, S. Kumaresan, Narosa Publishing House.

Note: 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	1	1	3	1	2	1	3	0	1
CO 2	3	3	1	1	3	1	3	1	3	0	1
CO 3	3	3	2	1	3	1	3	1	3	0	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓